Low-Bit Weight Quantization for Memory-Constrained LLM Serving

SYNTHETIC TEST DOCUMENT. All values in this document were invented to test the OmniResearch software and do not report real experiments.

Abstract

We study post-training weight quantization for serving a 13B-parameter decoder on a single GPU with 24 GB of VRAM. Four-bit group-wise quantization (group size 128) reduces weight memory from 26.0 GB to 7.4 GB while increasing WikiText-2 perplexity from 5.09 to 5.31. Three-bit quantization saves further memory but degrades perplexity to 6.12.

1 Introduction

Serving large language models on GPUs with limited VRAM is constrained primarily by weight memory and KV-cache memory. Weight-only quantization targets the first constraint. Activation outliers make 4-bit activation quantization difficult, so we quantize weights only and keep activations in FP16.

2 Method

Weights are quantized per output channel in groups of 128 using a calibration set of 128 sequences of 2048 tokens. Scales are stored in FP16. Dequantization is fused into the matrix-multiplication kernel.

3 Results

Table 1 reports memory and quality for the 13B model on an RTX 4090 (24 GB), batch size 1, sequence length 2048.

Precision	Weight memory (GB)	WikiText-2 PPL	Decode tokens/s
FP16	26.0	5.09	OOM
INT8 (W8A16)	13.5	5.11	38.2
INT4 g128 (W4A16)	7.4	5.31	61.7
INT3 g128 (W3A16)	5.8	6.12	64.0

Table 1: Memory, perplexity and decode throughput of the synthetic 13B model under weight quantization.

4 Discussion and Limitations

Four-bit quantization is the best trade-off in our setting: 3.5x lower weight memory for a 0.22 perplexity increase. Quantization does not reduce KV-cache memory, which dominates at long context lengths; KV-cache compression is complementary. We did not evaluate downstream task accuracy, so perplexity may understate quality loss.